Computer Organization (CSD206): Tutorial 5

Questions

1. Briefly define the elements of a machine instruction? What types of locations can hold source and destination operands?
2. Explain five important instruction set design issues.
3. Explain the range-extension rule for twos complement numbers?
4. Differentiate between pre-indexing and post-indexing addressing. Give an example in support of your answer.
5. What is a re-entrant procedure?
6. Write zero- and one-address machine instructions to compute X $=\frac{(A-B)}{(C-D*E)}$. Following instructions are available:
   For zero-address: PUSH M, POP M, ADD, MUL, SUB, and DIV.
   For one-address: LOAD M, STORE M, ADD M, SUB M, MUL M, DIV M.
7. Given x = 0101 and y = 1010 in twos complement notation, compute the product p = x * y with Booth's algorithm.
8. Divide -145 by 13 in binary twos complement notation, using 12-bit words. Verify your answer. [Hint: Use Dividend = Quotient * Divisor + Remainder]
9. A given microprocessor has words of 1 byte. What is the smallest and largest integer that can be represented in the following representations: Unsigned, Sign-magnitude, 1's complement, 2's complement?
10. Let the address stored in the program counter be designated by the symbol X1. The instruction stored in X1 has an address part (operand reference) X2. The operand needed to execute the instruction is stored in the memory word with address X3. An index register contains the value X4. What is the relationship between these various quantities if the addressing mode of the instruction is (a) direct; (b) indirect; (c) PC relative; (d) indexed?
11. Assume a stack-oriented processor that includes the stack operations PUSH and POP. Arithmetic operations automatically involve the top one or two stack elements. Begin with an empty stack. What stack elements remain after the following instructions are executed?

PUSH 4
PUSH 7
PUSH 8
ADD
PUSH 10
SUB
MUL

12. Given the following memory values and a one-address machine with an accumulator,
    - ✓ Word 20 contains 40
    - ✓ Word 30 contains 50.
    - ✓ Word 40 contains 60.
    - ✓ Word 50 contains 70.

    What values do the following instructions load into the accumulator?

    a. LOAD IMMEDIATE 20
    b. LOAD DIRECT 20
    c. LOAD INDIRECT 20
    d. LOAD IMMEDIATE 30
    e. LOAD DIRECT 30
    f. LOAD INDIRECT 30